LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DEEKSHITHA U
Register Number	192424055
Date	10 October 2026
Day	Saturday

SET 17: Employee Performance Analysis

Aim

To analyze employee performance using a line chart, bar chart, table, and an interactive Tableau dashboard.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

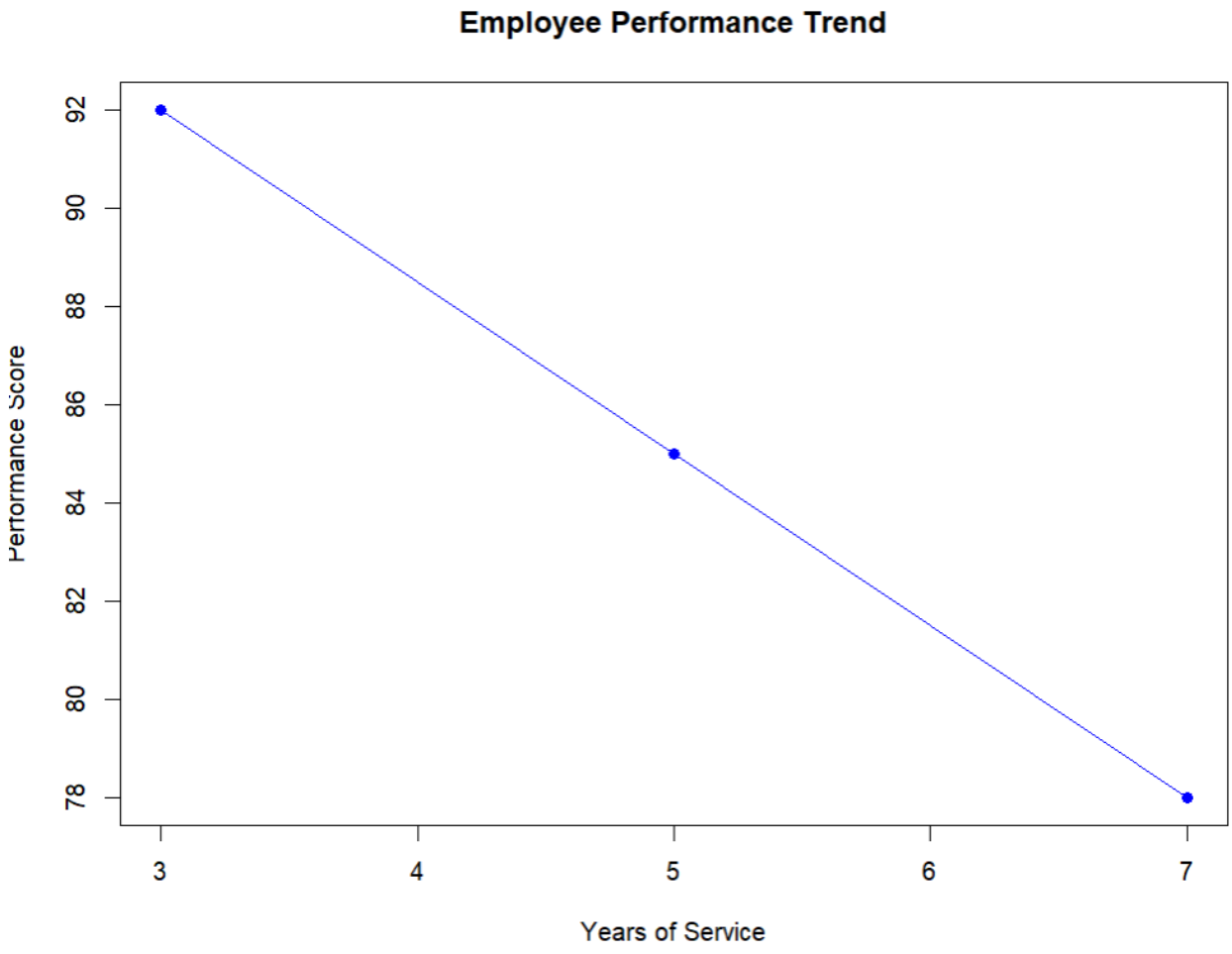
Q1. Create a line chart to visualize the performance trend of employees.

R Program

```
employee_id <- c(1, 2, 3)
years_service <- c(5, 3, 7)
performance <- c(85, 92, 78)
order_index <- order(years_service)
plot(
  years_service[order_index],
  performance[order_index],
  type = "o",
  pch = 19,
```

```
  xlab = "Years of Service",
  ylab = "Performance Score",
  main = "Employee Performance Trend",
  col = "blue"
)
```

Output:



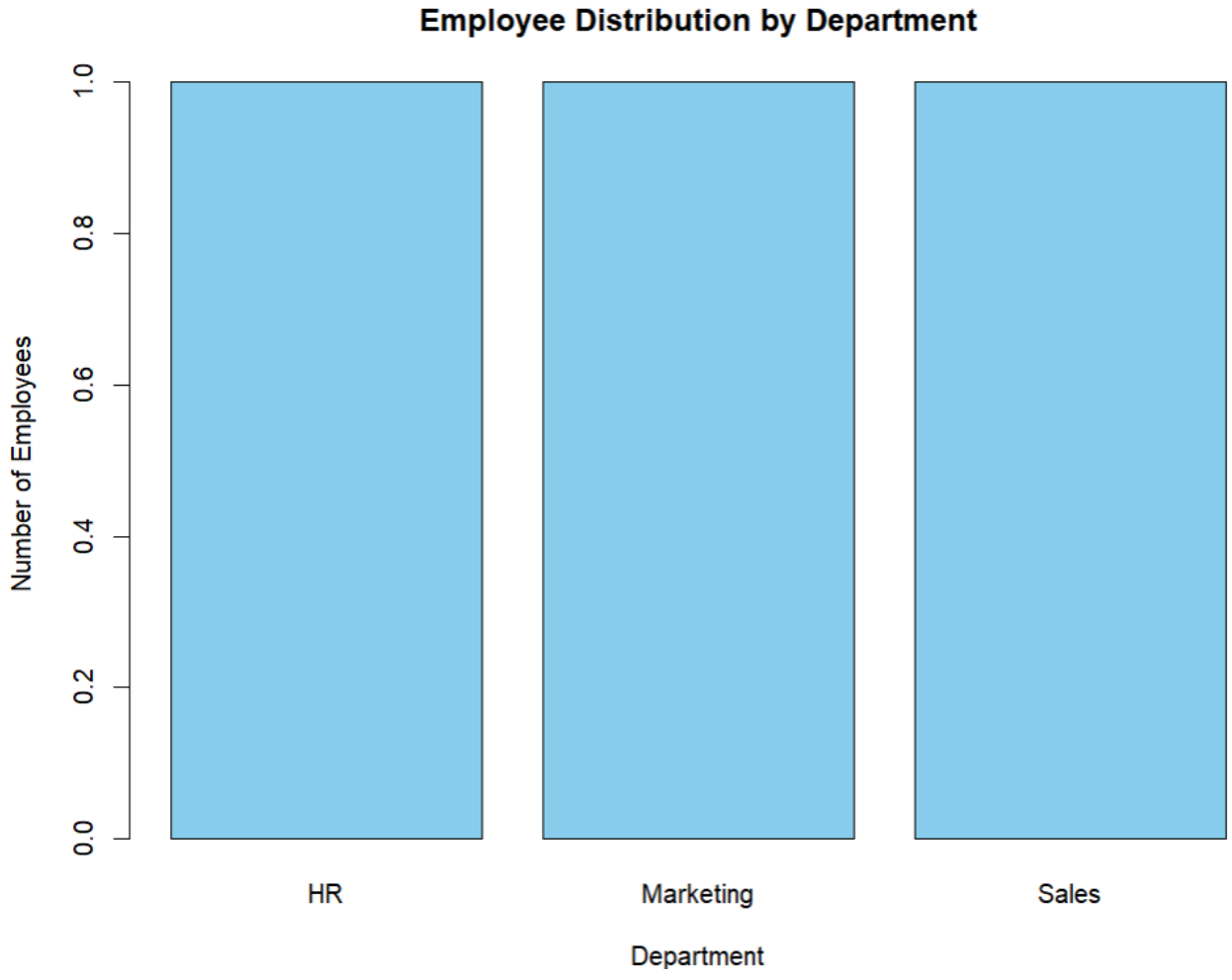
Observation: The employee with 3 years of service has the highest performance score (92), while the employee with 7 years has the lowest score (78).

Q2. Generate a bar chart showing the distribution of employees across departments.

R Program

```
department <- c("Sales", "HR", "Marketing")
department_count <- table(department)
barplot(
  department_count,
  names.arg = names(department_count),
  xlab = "Department",
  ylab = "Number of Employees",
  main = "Employee Distribution by Department",
  col = "skyblue"
)
```

Output:



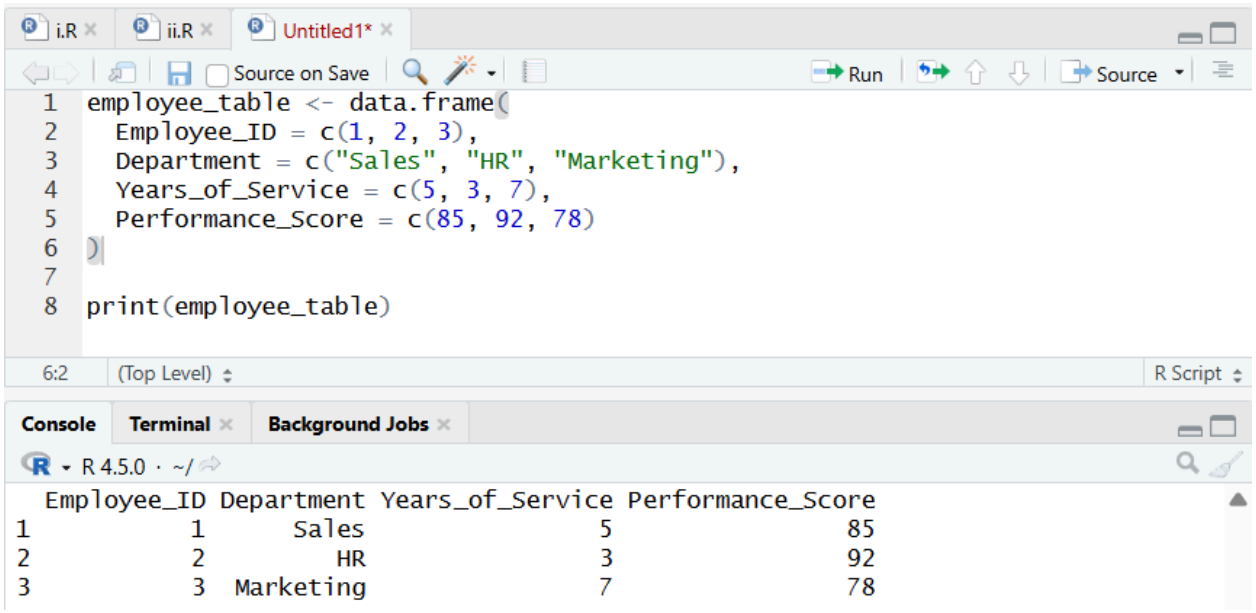
Observation: Sales, HR, and Marketing each have one employee.

Q3. Build a table to display the performance data for each employee.

R Program

```
employee_table <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
print(employee_table)
```

Output



Observation: The table displays the department, years of service, and performance score of each employee.

Q4. Develop a Tableau dashboard combining the line chart, bar chart, and table.

Step 1 — Prepare the CSV dataset

Save the following as Employee_Performance.csv.

Employee ID,Department,Years of Service,Performance Score
1,Sales,5,85
2,HR,3,92
3,Marketing,7,78

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Import Employee_Performance.csv.
4. Open a new worksheet.

Step 3 — Create the line chart

1. Create a new worksheet and rename it Performance Trend.
2. Drag Years of Service → Columns.
3. Drag Performance Score → Rows.
4. Select Line from the Marks dropdown.
5. Drag Employee ID → Label if you want to identify employees.
6. Set the title to Employee Performance Trend.

Step 4 — Create the bar chart

1. Create another worksheet named Department Distribution.
2. Drag Department → Columns.
3. Drag Employee ID → Rows.
4. Click the dropdown on Employee ID and select Measure → Count.
5. Select Bar from the Marks dropdown.
6. Drag CNT(Employee ID) → Label.
7. Set the title to Employee Distribution by Department.

Step 5 — Create the employee table

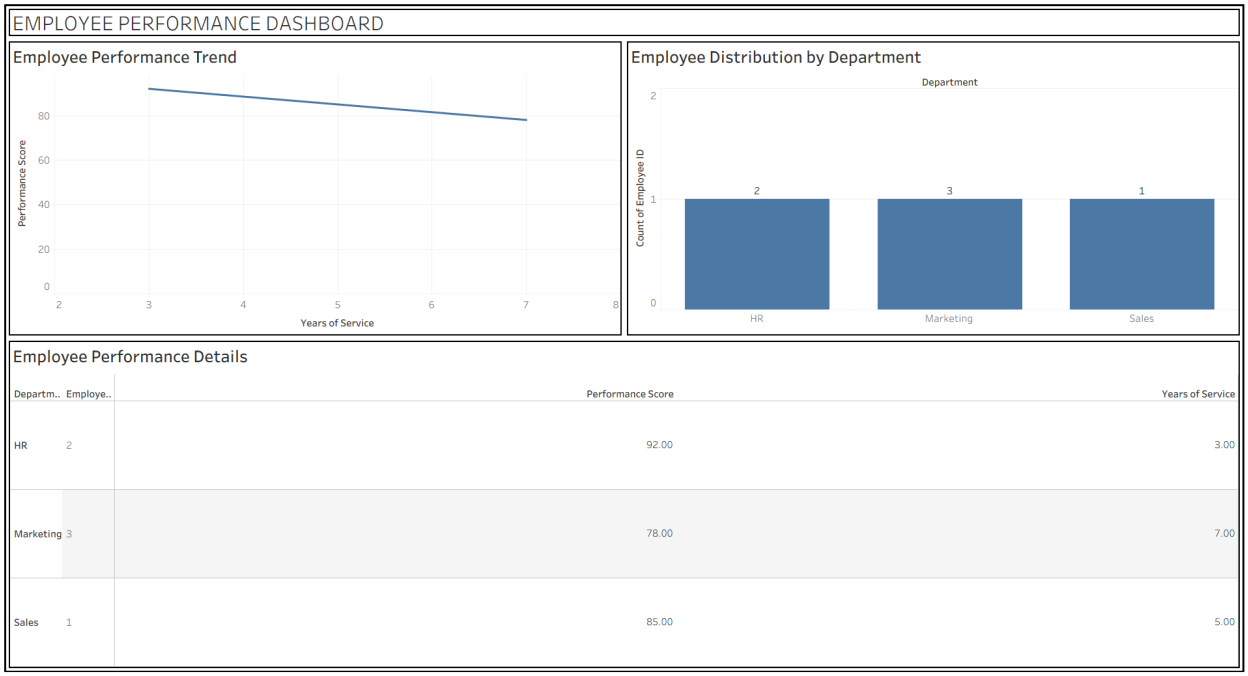
1. Create another worksheet named Employee Details.
2. Drag Employee ID → Rows.
3. Drag Department, Years of Service, and Performance Score into the view.
4. Select Show Me → Text Table if required.
5. Set the title to Employee Performance Details.

Step 6 — Create the dashboard

1. Click New Dashboard at the bottom.
2. Drag the Performance Trend worksheet to the top-left.
3. Drag the Department Distribution worksheet to the top-right.
4. Drag the Employee Details worksheet to the bottom.
5. Add filters such as Department if required.

6. Arrange the charts and table neatly and add the dashboard title Employee Performance Dashboard.

Expected dashboard



Procedure

1. Entered the employee performance dataset.
2. Created a line chart, bar chart, and table using R.
3. Imported the dataset into Tableau.
4. Created the three visualizations in separate worksheets.
5. Combined them into a dashboard for interactive exploration.

Result

The employee performance data was analyzed using charts and a table. A Tableau dashboard combining employee performance, department distribution, and employee details was designed successfully.